FREEDOM INTERNATIONAL SCHOOL

WORKSHEET-MCQ

PHYSICS

CURRENT ELECTRICITY

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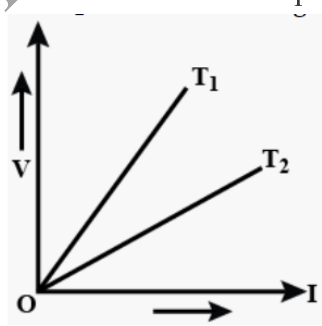
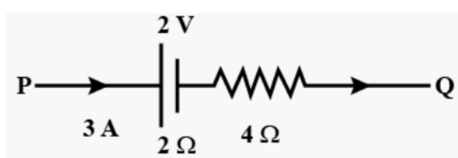
1. The current in a conductor varies with time t as $I = 2t+3t^2$, where I is in ampere and t in seconds. Electric charge flowing through a section of the conductor during t=2 s to t=3 s is
   (a) 10 C (b) 24 C (c) 33 C (d) 44 C
2. The ratio of masses of three wires is 1:2:3 and that of their lengths is 3:2:1. If the wires are made of same material, the ratio of their resistances will be
   (a) 1:1:1 (b) 1:2:3 (c) 9:4:1 (d) 27:6:1
3. The mobility of charge carriers increases with
   (a) increase in the average collision time (b) increase in the electric field
   (c) increase in the mass of the charge carriers (d) decrease in the charge of the mobile carriers
4. A wire has a resistance of 3.1 Ω at 30^0C and a resistance 4.5 Ω at 100^0C. The temperature coefficient of resistance of the wire is
   (a) $0.0064\,^0C^{-1}$ (b) $0.0025\,^0C^{-1}$ (c) $0.0034\,^0C^{-1}$ (d) $0.0012\,^0C^{-1}$
5. The voltage V versus current I graphs for a conductor at two different temperatures are shown in the figure. The relation between T_1 and T_2 is

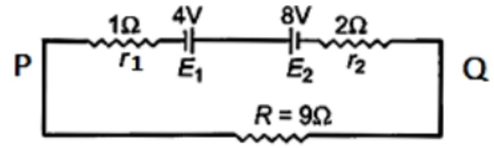


   (a) $T_1>T_2$ (b) $T_1<T_2$ (c) $T_1=T_2$ (d) $T_1=2\,T_2$
6. If 3 A of current is flowing between points P and Q in the circuit, then the potential difference between P and Q is



   (a) 30 V (b) 22 V (c) 20 V (d) 15 V
7. A cell supplies a current of 0.9 A through a 2 Ω resistor and a current of 0.3 A through a 7 Ω resistor. The internal resistance of the cell is
   (a) 2.0 Ω (b) 0.5 Ω (c) 1.0 Ω (d) 2.0 Ω

8. Two batteries of emf 4 V and 8 V with internal resistances 1 Ω and 2 Ω respectively are connected to an external resistance R= 9 Ω as shown in figure. The current in the circuit and the potential difference between P and Q respectively will be

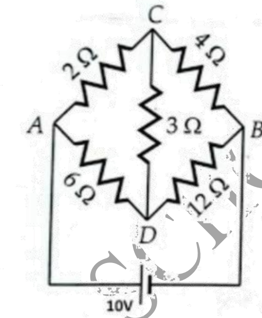


(a) 1/9 A, 9 V (b) 1/12 A, 12 V (c) 1/3 A, 3 V (d) 1/6 A, 4 V

9. The equivalent resistance (R_{AB}) between the points A and B is

(a) 6 Ω (b) 7.5 Ω

(c) 4.5 Ω (d) 8 Ω



10. Si and Cu are cooled from 300 K to a temperature of 60 K. Then resistivity

(a) for Si increases and for Cu decreases (b) for Cu increases and for Si decreases

(c) decreases for both Si and Cu (d) increases for both Si and Cu

For questions 11 to 15, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

11. **Assertion:** Though large number of free electrons are present in the metal, yet there is no current in the absence of electric field.
Reason: In the absence of electric field electrons move randomly in all directions.

12. **Assertion:** The value of temperature coefficient of resistance is positive for metals.
Reason: The temperature coefficient of resistance for insulator is also positive.

13. **Assertion:** The conductivity of an electrolyte is very low as compared to a metal at room temperature.
Reason: The number density of free ions in electrolyte is much smaller as compared to number density of free electrons in metals. Further, ions drift much slowly, being heavier.

14. **Assertion:** If the length of the conductor is doubled, the drift velocity will become half of the original value (keeping potential difference unchanged).
Reason: At constant potential difference drift velocity is inversely proportional to the length of the conductor.

15. **Assertion:** Terminal voltage of a cell is greater than e.m.f. of the cell, during charging of the cell.
Reason: The e.m.f. of a cell is always greater than its terminal voltage.